

Ibaad Muhammad

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EDUCATION

University College London

MEng Computer Science: First Class (Year 1)

Sep. 2025 – Jun. 2029

London, UK

British School Dhahran

A Levels: Mathematics A, Further Mathematics A*, Biology A, Chemistry A*

Aug. 2023 – Jun. 2025

Achieved highest mark in Computer Science AS Level in Eastern Province, Saudi Arabia

Khobar, Saudi Arabia

EXPERIENCE

Eternal Gems - Freelance Full-Stack Developer

Jul. 2026 – Present

- Designing and building a full-stack ecommerce platform (React, TypeScript, Node.js/Express, PostgreSQL) for a small business, implementing product catalogue, user authentication, basket, and Stripe payment integration
- Implemented secure user authentication using JWT and bcrypt password hashing, with httpOnly cookies for session storage and middleware-based route protection for authenticated pages
- Leveraging AI tools like Claude Code to brainstorm and verify, rather than generate code

Rolls-Royce - Novel Fission and CO₂ Reduction Project

Oct. 2025 – Jan. 2026

- Selected by Rolls-Royce and elected by peers to lead a team of 5 in constructing a Python-based high temperature gas reactor simulation model, projecting ~150 tonnes of CO₂ reduced daily
- Incorporated supervisor feedback from an interim presentation to adopt a structured, criteria-based scoring framework for evaluating design options, improving decision documentation across the project
- Validated model accuracy through automated unit testing, literature comparison, and programmatic graphing, achieving results within 1% of published values

PROJECTS

Habitly | Python, Flask, JavaScript, React, Tailwind CSS, SQLite, SQLAlchemy, Git

Jan. 2026 – Mar. 2026

- Created a full-stack activity recommendation web app with 3 coursemates using React, Flask, and SQLite
- Crafted a heuristic scoring algorithm combining user state inputs (physicality, sociability, time available) with time of day, recency bias and live weather data from the Open-Meteo API to tailor activity recommendations
- Architected the database schema and implemented secure user authentication with hashed password storage

Bioreactor IoT Control System | C++, JavaScript, Node-RED, JSON, MQTT, Git

Oct. 2025 – Dec. 2025

- Built a Node-RED/JavaScript dashboard with a monitoring page for live telemetry graphs and gauges, and a control page for setpoint and kill-switch commands, communicating with the rest of the system via MQTT
- Engineered C++ firmware for an ESP32, relaying JSON sensor telemetry and control commands between another ESP32 (via UART) and the MQTT broker with automatic WiFi reconnection handling
- Achieved closed-loop control of temperature, stirring, and pH with target accuracies of $\pm 1^\circ\text{C}$ (25–35°C), ± 10 RPM (100–1200 RPM), and ± 0.2 pH (pH 3–7)

Typocalypse | Unity, C#, Git

Sep. 2025 – Feb. 2026

- Developed a 2D cross-platform arcade game in Unity with C#, applying object-oriented programming principles, where players advance by typing to evade AI-controlled zombies, garnering hundreds of impressions
- Implemented online multiplayer using Unity Netcode with cloud matchmaking and relay services, debugging synchronisation issues including a race condition in the client joining sequence
- Designed the in-game art, sound effects and UI/UX to create a cohesive visual and audio experience

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, SQL, HTML, CSS, JavaScript, TypeScript

Frameworks/Libraries: Flask, React, Node.js, NumPy, pandas, Matplotlib, PostgreSQL, Express, pytest, SQLite

Developer Tools: Git, Docker, VS Code, PyCharm, IntelliJ IDEA, Claude Code, WSL, Unity, Vercel, Supabase

ACHIEVEMENTS & INTERESTS

- Published several original video games, totalling ~1000 impressions and generating revenue
- Awarded Gold and achieved highest score in school in the UKMT Senior Mathematical Challenge
- Paid Computer Science Outreach Ambassador at UCL, teaching younger students fundamental computing concepts through engaging workshops
- Selected for Expedia Group's Career Discovery Programme (2026), following CV screening and assessment
- Solved 350+ LeetCode problems in topics including trees, dynamic programming, graphs and sliding window
- Competed in multiple hackathons and game jams, prototyping functional applications under strict time constraints